

Student Utility Program

Java Lab Record

1. Aim

To develop a menu-driven Java Student Utility Program that accepts student details and marks, calculates the percentage and grade, and performs various operations using conditional statements, switch-case, loops, methods, and user input.

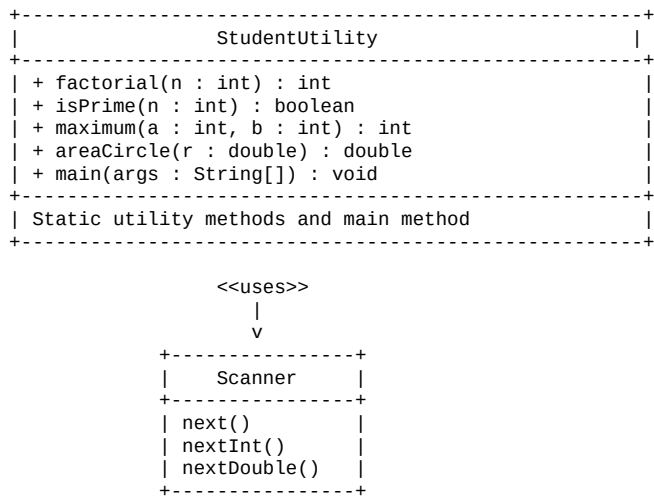
2. Step-by-Step Procedure

1. Create a Java class named StudentUtility.
2. Import the Scanner class to read input from the user.
3. Define methods for factorial, prime checking, maximum of two numbers, and area of a circle.
4. Display the student information heading and accept the student's name and roll number.
5. Read marks of five subjects and calculate the total marks.
6. Calculate the percentage and display the appropriate grade.
7. Display the menu containing Even/Odd, Largest of Three, Day of Week, Multiplication Table, Numbers from 1 to N, Sum of Natural Numbers, Fibonacci Series, Methods, and Exit.
8. Read the user's menu choice using Scanner.
9. Use switch-case to execute the selected operation.
10. Use a do-while loop to repeatedly display the menu until the user selects Exit.
11. Close the Scanner object after termination.

3. Algorithm

1. Start.
2. Create a Scanner object.
3. Display STUDENT INFORMATION.
4. Read student name and roll number.
5. Read marks of five subjects and calculate total marks.
6. Calculate percentage = total marks / 5.0.
7. Display student details, percentage, and grade.
8. Display the utility menu.
9. Read the user's choice.
10. If choice is 1, check whether the number is even or odd.
11. If choice is 2, find the largest of three numbers.
12. If choice is 3, display the day corresponding to the day number.
13. If choice is 4, display the multiplication table.
14. If choice is 5, display numbers from 1 to N.
15. If choice is 6, calculate the sum of the first N natural numbers.
16. If choice is 7, display the Fibonacci series.
17. If choice is 8, execute the methods for factorial, prime checking, maximum, and circle area.
18. If choice is 9, display Exit.
19. Otherwise, display Invalid Choice.
20. Repeat the menu until the choice is 9.
21. Close Scanner.
22. Stop.

4. UML Class Diagram



Main operations:

Student Details → Grade → Menu → Selected Operation

5. Code

```
package core_java;
import java.util.Scanner;

public class StudentUtility {

    // Factorial Method
    static int factorial(int n) {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact*i;
        }
        return fact;
    }

    // Prime Method
    static boolean isPrime(int n) {
        if (n <= 1)
            return false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    }

    // Maximum Method
    static int maximum(int a, int b) {
        if(a>b){
            return a;
        }else{
            return b;
        }
    }

    // Area of Circle Method
    static double areaCircle(double r) {
        A= 3.14*r*r;
        return A;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Part A: Student Details
        System.out.println(" STUDENT INFORMATION \n");

        System.out.print("Enter Student Name: ");
        String name = sc.next();

        System.out.print("Enter Roll Number: ");
        int roll = sc.nextInt();

        int total = 0;

        System.out.println("Enter Marks of 5 Subjects:");
        for (int i = 1; i <= 5; i++) {
            System.out.print("Subject " + i + ": ");
            total = total+sc.nextInt();
        }

        double percentage = total / 5.0;

        System.out.println("\nStudent Name : " + name);
        System.out.println("Roll Number : " + roll);
        System.out.println("Total Marks : " + total);
        System.out.println("Percentage : " + percentage);

        // Grade
        if (percentage >= 90)
            System.out.println("Grade : A+");
        else if (percentage >= 80)
            System.out.println("Grade : A");
        else if (percentage >= 70)
            System.out.println("Grade : B");
        else if (percentage >= 60)
            System.out.println("Grade : C");
        else
            System.out.println("Grade : Fail");

        int choice;

        do {

            System.out.println("1. Even or Odd");
            System.out.println("2. Largest of Three Numbers");
            System.out.println("3. Day of Week");
            System.out.println("4. Multiplication Table");
            System.out.println("5. Numbers from 1 to N");
            System.out.println("6. Sum of First N Natural Numbers");
            System.out.println("7. Fibonacci Series");
            System.out.println("8. Methods");
            System.out.println("9. Exit");

            System.out.print("Enter your choice: ");
```

```

choice = sc.nextInt();

switch (choice) {

    case 1:
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num % 2 == 0)
            System.out.println("Even Number");
        else
            System.out.println("Odd Number");
        break;

    case 2:
        System.out.print("Enter Three Numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if(a>b&&a>c)
            System.out.println("Largest Number = " + a);
        else if(b>a&&b>c)
            System.out.println("Largest Number = " + b);
        else
            System.out.println("Largest Number = " + c);
        break;

    case 3:
        System.out.print("Enter Day Number (1-7): ");
        int day = sc.nextInt();

        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            case 4:
                System.out.println("Thursday");
                break;
            case 5:
                System.out.println("Friday");
                break;
            case 6:
                System.out.println("Saturday");
                break;
            case 7:
                System.out.println("Sunday");
                break;
            default:
                System.out.println("Invalid Day");
        }
        break;

    case 4:
        System.out.print("Enter Number: ");
        int table = sc.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(table + " x " + i + " = " + (table * i));
        }
        break;

    case 5:
        System.out.print("Enter N: ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            System.out.print(i + " ");
        }
        break;

    case 6:
        System.out.print("Enter N: ");
        int num1 = sc.nextInt();

        int sum = 0;

        for (int i = 1; i <= num1; i++) {
            sum = sum + i;
        }

        System.out.println("Sum = " + sum);
        break;

    case 7:
        System.out.print("Enter Number of Terms: ");
        int terms = sc.nextInt();

        int x = 0, y = 1;

        System.out.print("Fibonacci Series: ");

        for (int i = 1; i <= terms; i++) {

```

```

        System.out.print(x + " ");
        int next = x + y;
        x = y;
        y = next;
    }
    break;
case 8:
    System.out.print("Enter Number for Factorial: ");
    int f = sc.nextInt();
    System.out.println("Factorial = " + factorial(f));

    System.out.print("Enter Number to Check Prime: ");
    int p = sc.nextInt();

    if (isPrime(p))
        System.out.println("Prime Number");
    else
        System.out.println("Not Prime");

    System.out.print("Enter Two Numbers: ");
    int m1 = sc.nextInt();
    int m2 = sc.nextInt();

    System.out.println("Maximum = " + maximum(m1, m2));

    System.out.print("Enter Radius: ");
    double r = sc.nextDouble();

    System.out.println("Area = " + areaCircle(r));

    break;
case 9:
    System.out.println("Exit");
    break;
default:
    System.out.println("Invalid Choice");
}
} while (choice != 9);
sc.close();
}
}

```

6. Sample Output

The following screenshots are included as the sample output/reference output for the Student Utility Program.

Output Screenshot 1

```
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac StudentUtility.java
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java StudentUtility
STUDENT INFORMATION

Enter Student Name: vyshu
Enter Roll Number: 166
Enter Marks of 5 Subjects:
Subject 1: 98
Subject 2: 97
Subject 3: 100
Subject 4: 94
Subject 5: 96

Student Name : vyshu
Roll Number : 166
Total Marks : 485
Percentage : 97.0
Grade : A+

1. Even or Odd
2. Largest of Three Numbers
3. Day of Week
4. Multiplication Table
5. Numbers from 1 to N
6. Sum of First N Natural Numbers
7. Fibonacci Series
8. Methods
9. Exit
Enter your choice: 1
Enter a number: 3
```

Output Screenshot 2

Odd Number

1. Even or Odd
2. Largest of Three Numbers
3. Day of Week
4. Multiplication Table
5. Numbers from 1 to N
6. Sum of First N Natural Numbers
7. Fibonacci Series
8. Methods
9. Exit

Enter your choice: 2

Enter Three Numbers: 98

97

34

Largest Number = 98

1. Even or Odd
2. Largest of Three Numbers
3. Day of Week
4. Multiplication Table
5. Numbers from 1 to N
6. Sum of First N Natural Numbers
7. Fibonacci Series
8. Methods
9. Exit

Enter your choice: 3

Enter Day Number (1-7): 5

Friday

Note: In the supplied code, the method `areaCircle()` uses `A` without declaring it. If you want to compile the program, declare a local variable such as `double A = 3.14*r*r;` before returning it. The code section above is kept as supplied.